UPAR Challenge 2027, Track 1: Pedestrian Attribute Recognition

Team: Muhammad Umar (23K-0023), Raghib Rizwan Rabbani (23K-0012), Syed Fahad Faheem Shah (23K-0062)

1. Task and approach

We took part in Track 1 of the UPAR Challenge 2027 (WACV 2027 Real-World Surveillance workshop) and finished with a Challenge_Avg of 0.7773 on the hidden test set, rank 42 on the leaderboard. Our best single model scored 0.7788. For every pedestrian crop the model outputs 40 probabilities: age, gender, hair length, clothing length and colour, lower-body type and accessories. The scorer cuts them at 0.5 and ranks teams by the harmonic mean of mA and instance F1, computed on each test dataset. Training is limited to the organisers' data (Market1501, PA-100K, PETA), while the 29,248 test images come from an unseen domain. Our final submission averages two CLIP ViT-L/14 networks fine-tuned with one sigmoid per attribute.

2. What the data showed

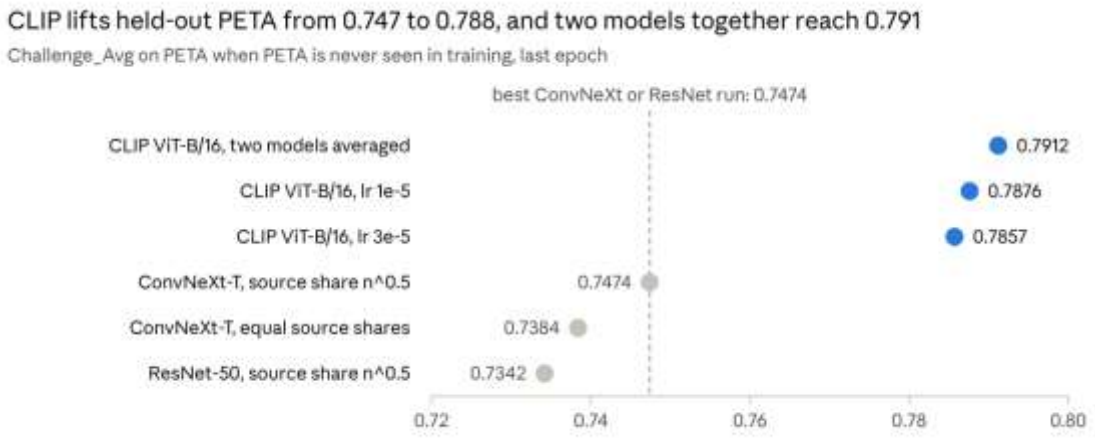
- **The sources differ a lot.** PA-100K is 81% of training but Market1501 is 49% of validation, so we always scored each source separately, as the leaderboard does.
- **Rare and missing labels.** Twelve attributes are positive in under 2% of training images (30% of mA). Five colour attributes are never annotated in Market1501 and one in PETA, so we masked those zeros out of the loss.
- **Labels are not one-hot.** About 11 to 12% of PETA images have two upper-body colours, so we used independent sigmoids. The crops are tiny (Market1501 is always 64x128).

3. Method

1. **Cache.** Crops are resized to 192x96 keeping the aspect ratio, padded, and stored once as uint8, because Colab gives us about two CPU cores.
2. **GPU augmentation.** Flip, small rotation and zoom, brightness, contrast and saturation changes, low-resolution blur and random erasing. We never change hue, since colours are labels.
3. **Loss and sampling.** Binary cross-entropy with positive weights capped at 10, masked for never-annotated attributes, and source shares proportional to size to the power 0.5. AdamW with a cosine schedule, fp16, 3 to 8 epochs.
4. **Backbone.** ConvNeXt-Tiny first, then CLIP ViT-B/16 and finally ViT-L/14 (OpenAI weights, 196x98 input for L/14) with layer-wise learning-rate decay. The final models are trained on train plus validation, averaged in logit space and cut at 0.5. We ship a plain-PyTorch copy of the CLIP tower, so the package needs only torch, torchvision, numpy and Pillow.

4. Experiments and results

Switching from ConvNeXt-Tiny to CLIP was the one change that clearly mattered. We compared runs by training on two sources and scoring the third, our stand-in for the unseen test domain.



Source: our held-out-source training logs, last epoch (8 epochs for ConvNeXt and ResNet, 5 for CLIP); single seed

- **Shorter training won.** A 34-epoch ConvNeXt run peaked on validation at epoch 4, while an 8-epoch schedule reached 0.853.
- **Small extras.** Flip test-time augmentation added 0.001 and a second CLIP model about 0.004. Threshold tuning on held-out logits did not transfer, so we kept the plain 0.5 cut.

Submission	Model	mA	Instance_F1	Challenge_Avg	Rank
1	Sample baseline (attribute prior)	0.5000	0.4910	0.4955	70
2	ConvNeXt-Tiny, train+val	0.7302	0.7274	0.7288	60
3	CLIP ViT-B/16, one model	0.7693	0.7642	0.7668	44
4	Three CLIP ViT-B/16 models (final)	0.7676	0.7713	0.7694	41
5	CLIP ViT-L/14, one model (best single)	0.7802	0.7774	0.7788	41
6	Two CLIP ViT-L/14 models (final)	0.7754	0.7792	0.7773	42

The jump from B/16 to L/14 (+0.0094 as a single model against the three-model B/16 ensemble) came from a larger backbone. Averaging two L/14 models (plain fine-tuning and linear-probe-then-fine-tune) gave 0.7773, which is within noise of the single model. That ensemble used the last allowed submission, so 0.7773 is our final score. The top leaderboard score was 0.8021 when we finished.

5. Challenges

- **Domain shift.** ConvNeXt scored 0.853 on validation but 0.7288 on the test set. CLIP's more robust features recovered most of that gap.
- **Rare attributes under a fixed 0.5 cut.** Capped positive weights helped, while threshold tuning did not.
- **Free Colab.** About two CPU cores, 6 GB of Drive and 1 GB checkpoints. The cache, GPU augmentation, resumable checkpoints and three accounts running in parallel kept it workable.
- **Matching training and submission.** The packaged run.py reproduced the training scores (0.8529 against 0.8528).

6. Possible improvements

- Longer training and more seeds for ViT-L/14, which only got 3 to 4 epochs at 196x98 on a T4.
- Stronger blur and low-resolution augmentation aimed at the PETA gap, and CLIP text prompts as attribute classifiers.
- More seeds and several held-out folds, to separate small gains from noise.

7. Proof of submission and rules

Final submission: Challenge_Avg 0.7773 (mA 0.7754, Instance_F1 0.7792), submission ID 1010210, the two-model L/14 ensemble. Our best single model was ID 1006798 at 0.7788. Both are shown in the attached screenshots. We used only the organisers' data, ImageNet and OpenAI CLIP weights, no test images for training or tuning, and run.py never uses image names or folders.

ID #	File name	Date	Status	Score	Actions
1010210	06_submission_l14_ensemble.zip	2026-10-26 22:39	Finished	0.7773	
1006798	submission_l14.zip	2026-10-21 06:27	Finished	0.7788	
1003348	submission_clip_ens.zip	2026-10-19 21:01	Failed	0.7694	
1002923	submission_clip.zip	2026-10-19 17:25	Finished	0.7668	
1001777	submission_final8.zip	2026-10-19 05:45	Finished	0.7288	
1001365	submission_baseline.zip	2026-10-18 23:16	Finished	0.4955	

My Submissions

41	yunseon	2026-09-28 09:51	949141	949141_VID-Docker-GPU	**0.7782**	0.7803	0.5555	0.6601	0.7414	0.8295	0.776
42	m_umar	2026-10-26 22:39	1010210	1010210_VID-Docker-GPU	**0.7773**	0.7754	0.5812	0.6638	0.7447	0.8319	0.7792
43	nhm_giang	2026-10-14 16:51	987374	987374_VID-Docker-GPU	**0.776**	0.7775	0.5798	0.66	0.7565	0.8068	0.7745

Leaderboard